「Def. A differentiable function $f: \mathbb{R}^n \to \mathbb{R}^m$ is called a mapping.

「Def. Let $F: \mathbb{R}^n \to \mathbb{R}^m$ be a mapping and V_p be a tangent vector.
Then, for a curve $\beta: (-\varepsilon, \varepsilon) \to \mathbb{R}^m : t \mapsto F(p + t \cdot v)$, define

$$F_*(V_p) \overset{def}{=} \beta'(0)_{F(p)}.$$

The map $F_*: \bigcup_{p \in \mathbb{R}^3} T_p \mathbb{R}^3 \to \bigcup_{p \in \mathbb{R}^3} T_{F(p)} \mathbb{R}^3$ is called the tangent map of F.

「Lemma. Let $F = (f_1, \cdots, f_m): \mathbb{R}^n \to \mathbb{R}^m$ be a mapping. Given a tangent vector V_p,
Proof.

$$F_*(V_p) = (V_p[f_1], V_p[f_2], \cdots, V_p[f_m])$$

$$= (Df_1(p) \cdot v, Df_2(p) \cdot v, \cdots, Df_m(p) \cdot v)$$

$$= DF(p) \cdot v$$

where $DF(p)$ is the Jacobian, $DF(p) = \left(\frac{\partial f_i}{\partial x_j}(p) \right)_{ij}$.

Note: For fixed $p \in \mathbb{R}^3$, F_{*p} is linear with respect to the vector v at p.

「Def. A mapping $F: \mathbb{R}^n \to \mathbb{R}^m$ is called regular if: for each $p \in \mathbb{R}^3$, F_{*p} is injective.